

Real Time Face Attendance System

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Contents :

Abstract

1.Introduction

1. Problem Definition & Description

2. Objective of the Project

- **Aim of the project**

1.3 Scope of the Project

- **Determining goals**
- **Data & Constrains**
- **Workflow management strategies**

2.System Analysis

1. Existing System

- **Background & Literature Survey**
- **Limitations of Existing System**

2. Proposed System

- **Advantages of Proposed System**

3. Software & Hardware Requirements

- **Software Requirements**
- **Hardware Requirements**

4. Feasibility Study

- **Technical Feasibility**
- **Robustness & Reliability**
- **Economic Feasibility**

3.Architectural Design

1. Module Design

- 1. User Management Module**
- 2. Authentication module**
- 3. Risk Assessment Module**
- 4. Logging and monitoring module**

2. Method & Algorithm design

3. 1. Algorithm

4. Project Architecture

- 1. Architectural Diagram**
- 2. Class Diagram**
- 3. Use case Diagram**
- 4. Sequence Diagram**
- 5. Feature Extraction**

4.Implementation &Testing

- 1. Coding Blocks**
- 2. Sample Code**
- 3. Execution Flow**
 - 1. Main frame of the System**
 - 2. Connect to the DataBase**
 - 3. Taking images for the Students**
 - 4. Choose the Subject and Fill Attendance**
 - 5. Train the Model**
- 4. Testing**
 - 1. User Interface**
 - 2. Enter the roll number**
 - 3. Enter the subject**
 - 4. Detecting the face**
 - 5. Marking them Present**

Abstract:

One's face is a physical depiction of who they are. As a result, we have suggested a facial recognition-based automatic student attendance system. The usage of face recognition technology in everyday life is particularly beneficial for security control systems. Face recognition is used by the FBI (Federal Bureau of Investigation) for criminal investigations as well as the airport security system to identify criminals. In the first step of our suggested method, video framing, the camera is activated using a user-friendly interface. The Viola-Jones method is used to identify and segment the face ROI from the video frame. If necessary, image size scaling is done at the pre-processing step to prevent information loss. After using the median filter to reduce noise, color pictures are transformed into grayscale versions. Then, to improve the contrast of the pictures, contrast-limited adaptive histogram equalization (CLAHE) is applied. Principal component analysis (PCA) and improved local binary pattern (LBP) are used appropriately in the face identification stage to extract the characteristics from facial pictures. By minimizing the lighting impact and boosting identification rates, the improved local binary pattern outperforms the original local binary pattern in our suggested method. The features that were retrieved from the training pictures and the features that were extracted from the test images are then compared. The face pictures are then identified and categorized using the upgraded LBP, PCA, and algorithm that produced the best results. The acknowledged student's attendance will then be recorded and saved in the excel file. A warning will be sent if a student signs in more than once. The student who is not enrolled will also be allowed to register right away. When two photographs per person are trained, the average recognition accuracy is 100% for high-quality images, 94.12% for low-quality images, and 95.76% for the Yale face database.

1. Introduction:

1.1 problem Definition &Description

• Problem Definition

In educational institutions, accurately tracking student attendance is crucial for managing class participation, academic performance, and administrative reporting. Traditional methods, such as manual roll calls and paper-based systems, are time-consuming, prone to human error, and susceptible to manipulation. There is a need for an automated, efficient, and reliable system to streamline the attendance process and improve accuracy and security.

• Project Description

This project aims to develop a user-friendly, automated student attendance system using face recognition technology implemented through a MATLAB GUI. The system employs an enhanced Local Binary Pattern (LBP) method with a radius of two for face recognition. The choice of radius size is analyzed and discussed to optimize performance. By automating attendance tracking, the system aims to reduce administrative workload, eliminate errors associated with manual processes, and ensure a secure and accurate record of student attendance.

1.2 Objectives of the Project

.Aim of the project

The goal of this project is to create an automated student attendance system based on facial recognition. The following are expected results to meet the objectives:

To identify the facial segment in the video frame.

- To identify the useful aspects on the observed face.
- To categorize the traits in order to identify the observed face.
- To note the indicated student's Tasks and deliverables

1.3 Scope of the project

- **Determining Goals**

The primary goal of this project is to automate student attendance using facial recognition technology, thereby eliminating the need for manual roll calls and paper-based records. This includes developing a user-friendly MATLAB GUI that is easy to navigate for both students and administrators, featuring functionalities such as student registration, attendance tracking, and system updates. Ensuring the system's accuracy and reliability is crucial, necessitating robust algorithms capable of handling variations in lighting, pose, and facial expressions. Real-time processing capabilities are essential to provide immediate and efficient attendance recording, optimized to manage large numbers of students swiftly. Additionally, the project prioritizes security and privacy, with protocols to protect stored data and prevent unauthorized access and spoofing attempts.

- **Data & Constraints**

Data requirements for the project include high-quality facial images for student enrollment and recognition, a secure database for storing images and attendance records, and labeled datasets for training and testing the facial recognition algorithm. Technical constraints involve ensuring the system operates efficiently within the computational capabilities of available hardware and optimizing algorithms for real-time processing. Environmental constraints include managing variations in classroom lighting, backgrounds, and camera angles, as well as changes in students' appearances over time. Legal and ethical constraints necessitate compliance with data protection laws and ethical considerations in the use of facial recognition technology, such as obtaining consent and ensuring transparency. User constraints focus on creating a system that is easy to use for non-technical users, providing clear instructions and support for registration and interaction with the system.

2. System Analysis

2.1 Existing System

- **Background & Literature Survey**

To recognize relatives, friends, or other people we are familiar with, face recognition is essential in daily life. We might not realize that several steps have been taken to recognize human faces. Our ability to acquire information and analyze it throughout the recognition process is a result of human intellect. The image that is transmitted to our eyes, and more particularly, the retina, provides us with information in the form of light. Electromagnetic waves take the shape of light, which is projected to human eyesight after being transmitted from a source onto an object. According to Robinson-Riegler, G., & Robinson-Riegler, B. (2008), when the human visual system has processed the image, we really identify the object's shape, size, contour, and texture in order to analyze the data. The analysed information will be compared to other representations of objects or face that exist in our memory to recognize.

• **Limitations of Existing System**

The existing systems for student attendance tracking have several limitations. Manual roll calls and paper-based registers are time-consuming and prone to human error, consuming valuable class time and often resulting in incorrect or missed entries. Semi-automated systems, such as swipe cards and RFID tags, while faster, still require physical interaction, leading to delays and potential issues like lost or forgotten cards. These systems can also be manipulated, allowing for proxy attendance where one student swipes in for another. Furthermore, existing facial recognition systems, primarily used for security, are often complex, expensive, and not optimized for educational settings. They lack integration with school management systems, making it difficult to maintain comprehensive and accurate attendance records. These limitations highlight the need for a more efficient, reliable, and user-friendly solution tailored to the needs of educational institutions.

2.2 Proposed System

The RFID (Radio Frequency Identification) card system, the fingerprint system, and the iris recognition system all have drawbacks, according to Arun Katara et al. (2017). The RFID card method is used because it is straightforward. If the user has their friend's ID card, they usually assist their pals with checking in. The fingerprint technology is efficient but not very quick because each user must stand in line and go through the verification procedure individually. The human face, however, is constantly visible and contains less data than the iris when used for facial identification. A more detailed iris recognition technology could violate the user's privacy. Although voice recognition is a possibility, it is not as precise as other approaches. Thus, it is advised that the student attendance system be upgraded to include face recognition technology.

2.3 Software and Hardware Requirements

- **Software Requirements**

1. Backend: Django (Python) REST framework.
2. MicroSoft Excel

- **Hardware Requirements**

1. CPU: Quad core or higher
2. RAM: Aim between 8gb -16gb
3. Storage: Minimum 50GB
4. Camera:Minimun16PX

2.4 Feasibility Study

1. Technical Feasibility

The technical feasibility of the face recognition student attendance system is high. Advances in computer vision and machine learning have made facial recognition technology robust and accurate. MATLAB provides a powerful platform for developing such applications, offering extensive libraries and toolkits for image processing and GUI development. The system can be implemented on standard computer hardware with a camera, making it accessible for most educational institutions. Additionally, the use of Local Binary Patterns (LBP) and other efficient algorithms ensures real-time processing and high accuracy, even with variations in lighting and facial expressions.

2. Economic Feasibility

Economically, the project is feasible for most educational institutions. The primary costs involve acquiring or upgrading existing hardware, such as cameras and computers, and purchasing MATLAB licenses if not already available. Once implemented, the system reduces the time and labor costs associated with manual attendance tracking, providing long-term savings. The benefits of increased accuracy and reduced administrative workload further justify the initial investment. Additionally, as the technology matures and becomes more widespread, the costs of hardware and software are expected to decrease.

3.Architectural design

3.1 Module Design

3.1.1 User Management Module:

It automates the process of capturing and monitoring student attendance through facial recognition, eliminating manual roll calls. The system manages the database of student information, conducts face detection and recognition, and provides a user-friendly interface for interaction. Additionally, it offers data analysis and reporting functionalities, ensuring accurate attendance tracking and facilitating administrative decision-making. Through robust security measures, the system maintains access control and safeguards student data privacy, enhancing overall efficiency and accountability in attendance management.

3.1.2 Authentication Module:

This Module manages user access to the face recognition student attendance system. It verifies user identities through login credentials or biometric authentication. Upon successful authentication, users gain access to system functionalities based on their assigned roles and permissions. The module ensures data security and integrity by enforcing authentication protocols and access control measures, thereby maintaining the confidentiality of student information and system resources.

3.1.3 Risk Assessment Module:

The Risk Assessment Module identifies and evaluates potential risks associated with the face recognition student attendance system. It assesses factors such as technical failures, data breaches, privacy concerns, and system vulnerabilities. By analyzing these risks, the module helps develop mitigation strategies to minimize the impact of adverse events on system functionality and security. Additionally, it facilitates proactive risk management practices, ensuring the system's reliability, integrity, and compliance with regulatory requirements.

3.1.4 Logging and Monitoring Module:

The face recognition student attendance system is responsible for tracking system activities and generating comprehensive logs for auditing and analysis. It records user authentication events, attendance tracking activities, database modifications, and system errors. These logs provide administrators with insights into system usage, security breaches, and performance issues, enabling timely detection and resolution of issues. Additionally, the module includes real-time monitoring capabilities to ensure system functionality and integrity, allowing for proactive management and optimization of the attendance system.

3.2 Methodology &Algorithm

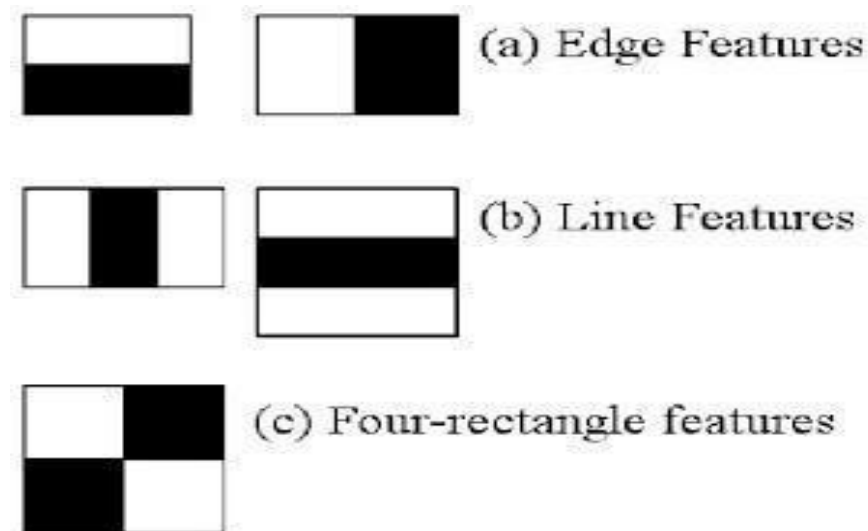
The method uses a facial recognition-based attendance system for students. The technique starts with the picture capturing utilizing an easy-to-use interface, then moves on to pre-processing the facial photos that were collected, feature extraction from those images, subjective selection, and finally categorization of the facial images to be recognized. This suggested technique computes both LBP and PCA feature extraction algorithms, then compares their performance. With this method, LBP is improved to lessen the lighting impact. Moreover, a subjective selection method that combines improved LBP and PCA is created to improve accuracy. The next sections will go through the specifics of each level.

The proposed system's flow chart is divided into two sections, The first section is for training pictures, and the second is for testing images.

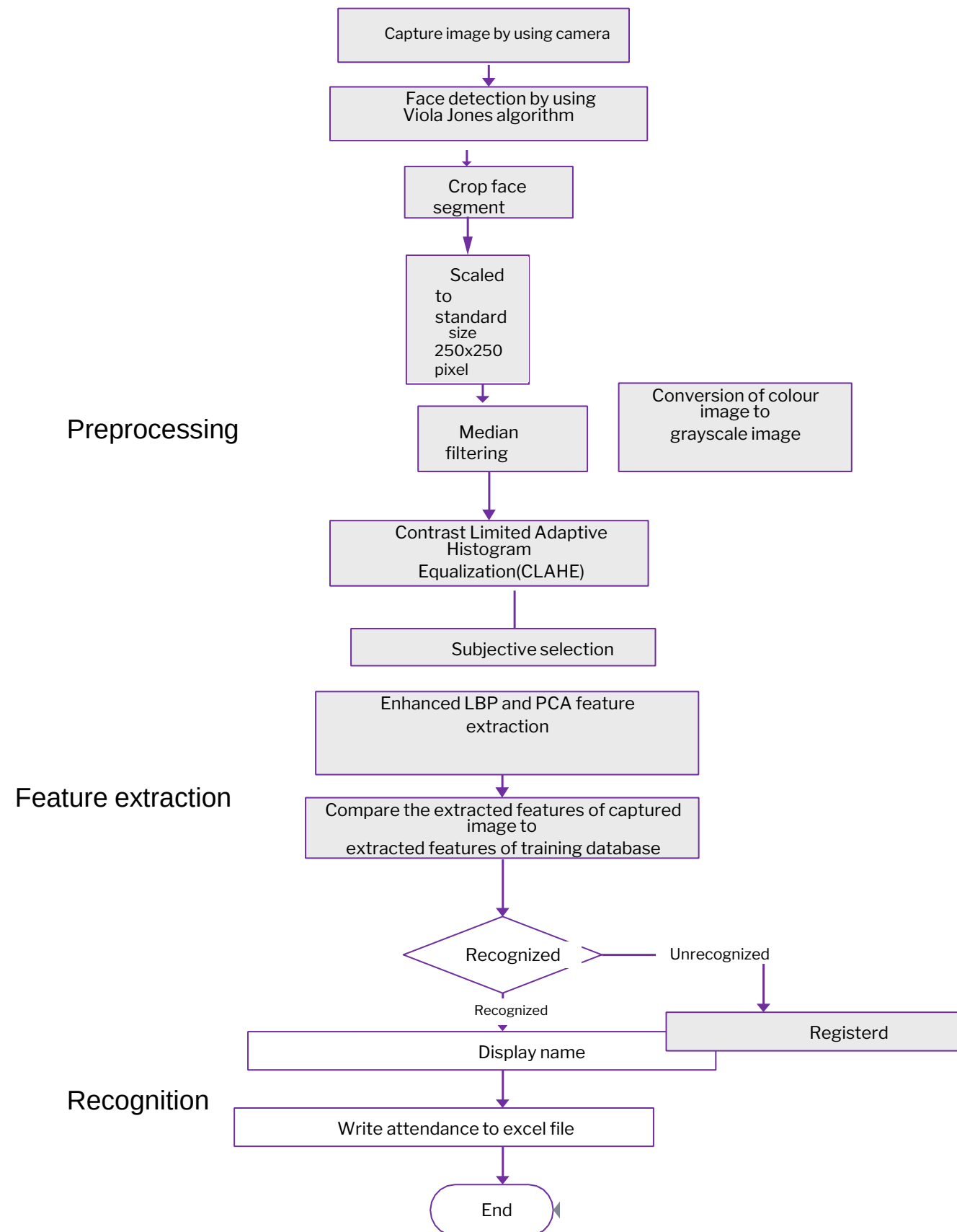
Viola-JonesAlgorithm:

The most widely used technique to locate the face segment from a static picture or video frame is the Viola-Jones algorithm, which was developed by P. Viola and M. J. Jones in 2001. The Viola-Jones algorithm is essentially composed of four pieces. The first component is called the Haar feature, the second component is where the integral picture is formed, the third component is where Adaboost is implemented, and the fourth component is the cascade process.

Figure 1.1



3.3.1 Architecture Diagram



3.3.3 Class Diagram

Figure 2.1

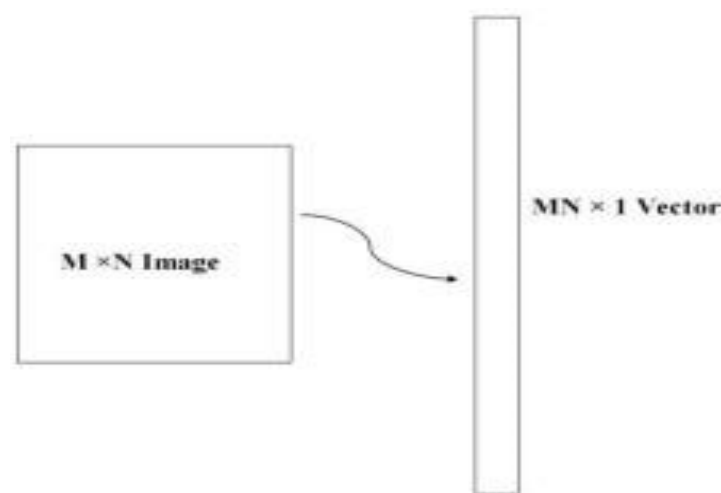
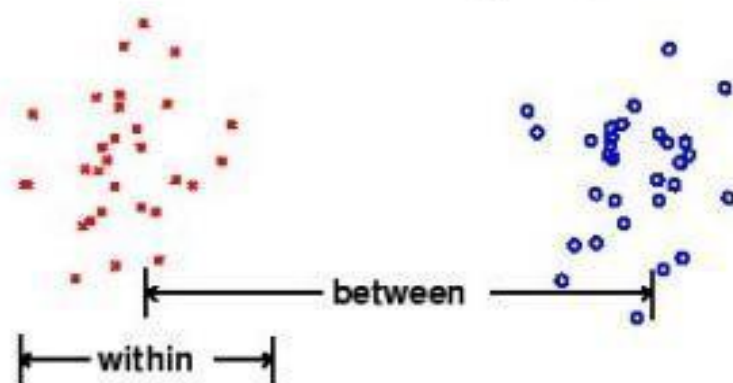


Figure 2.2

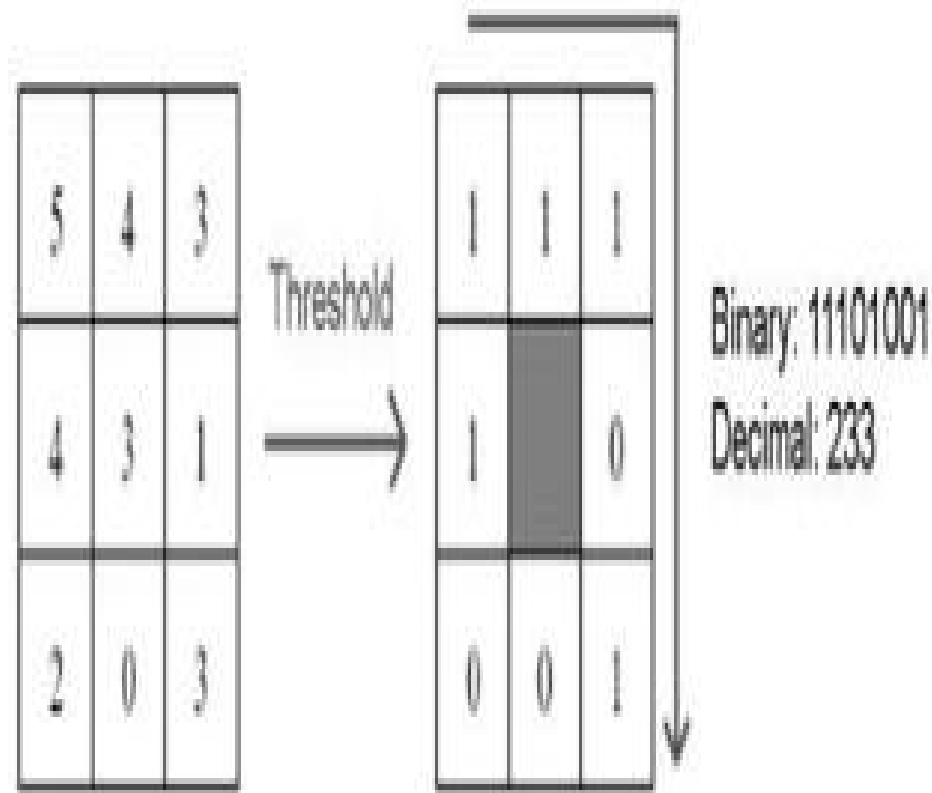
Good class separation



A feature is a collection of data that symbolizes the content of a picture. The most important step in face recognition is facial feature extraction. Yet, choosing features might be difficult. For a feature extraction method to produce results with high accuracy, it must be consistent and stable throughout a range of changes. For face recognition, there are a few feature extraction techniques. Abhishek Singh and Saurabh Kumar (2012), Liton Chandra Paul and Abdulla Al Sumam (2012), and Bhuvaneshwari et al. (2017) suggested PCA for face recognition. PCA was also employed by D. Nithya (2015) in a facial recognition-based student attendance system. PCA is renowned for its accurate and quick calculation. In essence, PCA keeps the data's variety and eliminates pointless existing connections between the original components.

3.3.4 Use Case Diagram

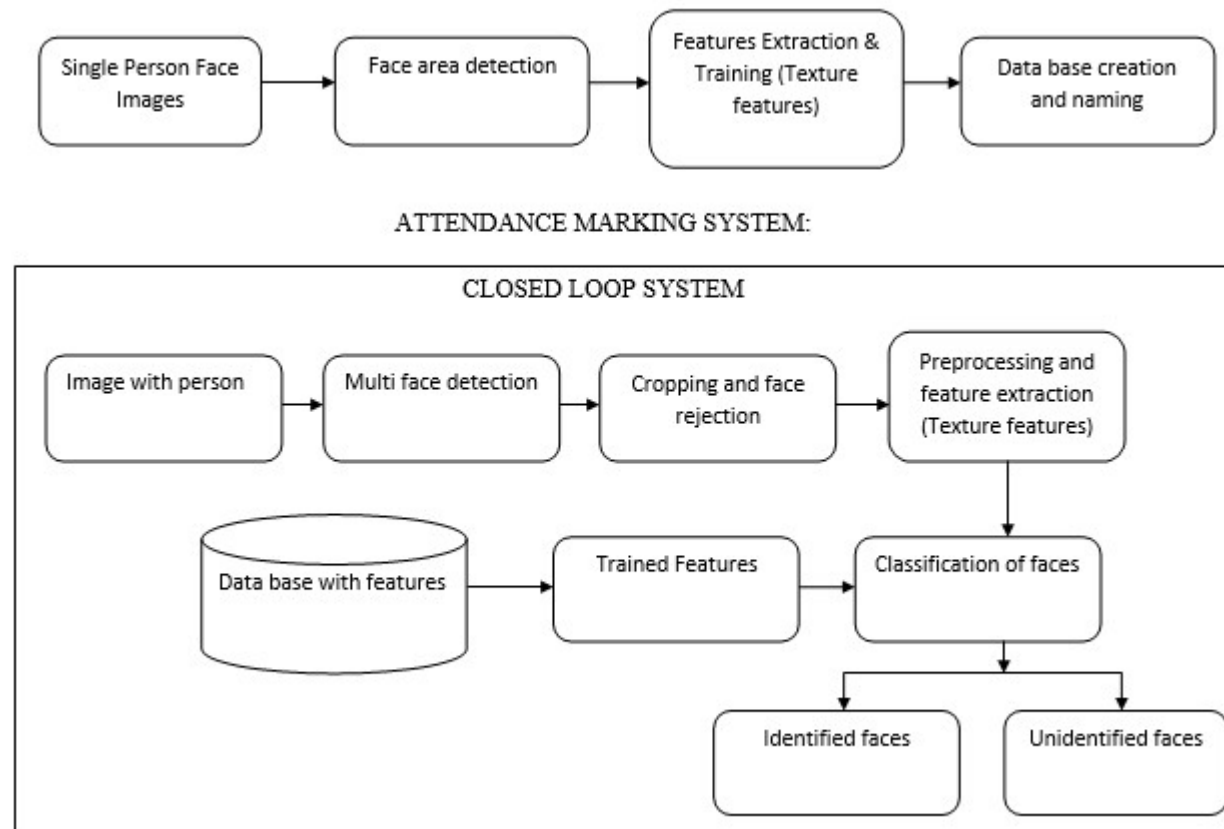
Figure 3.1



- The study by Timo Ojala et al. introduced the initial LBP (Local Binary Patterns) operator (2002). In their 2013 study, Md. Abdur Rahim et al. suggested using LBP to extract texture information as well as contour information to depict facial pictures. Each face picture is divided into smaller parts using LBP, and the histogram for each region is then retrieved. Each region's histograms are combined into a single feature vector. The facial picture is represented by this feature vector, and the Chi square statistic is used to assess the degree of similarity across face images. Each region's smallest window measures three by three pixels. It is calculated by thresholding each pixel in a window, with the threshold value being the middle pixel. Neighborhoods with values above the threshold are given a 1 while those with values below the threshold are given a 0. The center pixel will then be represented by a byte value created from the resultant binary pixels.

3.3.5 Sequence Diagram

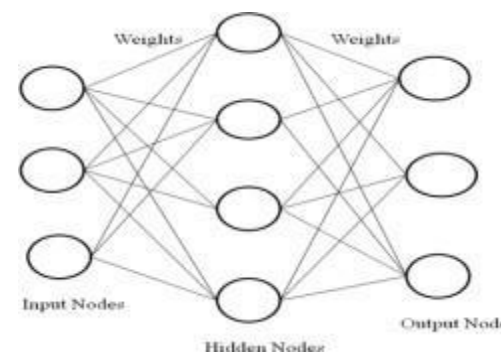
Figure 3.2



Classification involves the process of identification of faces. The distance classifier finds the distance between the test image and train image based on the extracted features. The smaller the distance between the input feature points and the trained feature points, the higher the similarity of the test image and training image. In other words, the facial images with the smallest/minimum distance will be classified as the same person. Deepesh Raj (2011) mentioned several types of distance classifiers such as Euclidean Distance, City Block Distance and Mahalanobis distance for face recognition. Md. Abdur Rahim et al. (2013) implemented Chi-Square statistics as distance classifier for LBP operators.

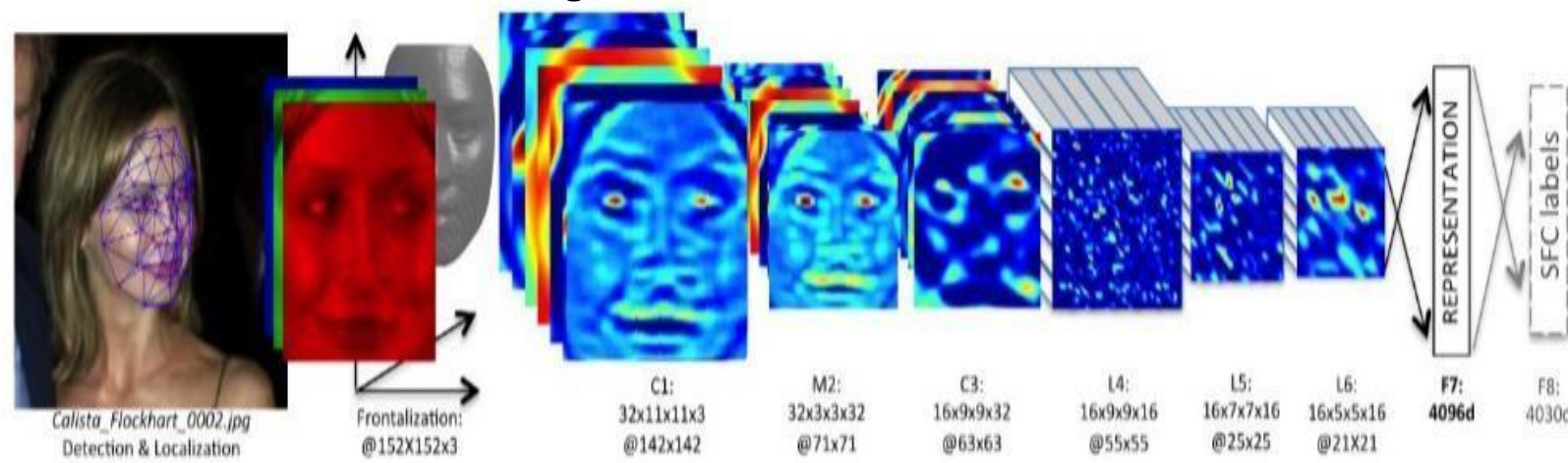
- Many benefits of LBP make it attractive to adopt. It can handle a wide range of face expressions, picture rotation, and human ageing and has a high tolerance for monotonic light fluctuations. Because of these dominating qualities, LBP is frequently used in real-time applications. Initially, face identification is the only use of neural networks. It is then researched further to be used for facial recognition. Manisha M. Kasar et al. (2016) examined artificial neural networks (ANN) for facial recognition in their article. The "nodes" of an ANN are a network of synthetic neurons. For the purposes of categorization and recognition, the nodes simulate the human brain. The strength of the connections between these nodes is determined by the values that are assigned to each of their connections. Strong relationship is indicated by a high value. Input nodes, hidden nodes, and output nodes are the three types of nodes or layers that make up a neuron. Depending on its influence, input nodes are given weight. Certain mathematical functions and thresholding functions are used in hidden nodes to make predictions or probabilities, which are then used to block superfluous inputs and produce output nodes. More than one layer can make up a hidden node. In the output node, several inputs result in a single output.

Figure 3.3



- Another neural network approach for face recognition is the convolutional neural network (CNN). CNN also has an input layer, a hidden layer, and an output layer. Convolutional, pooling, fully connected, and normalizing layers are among the layers that make up a CNN's hidden layers. For CNN to function effectively, however, thousands or millions of facial photos must be taught, and this training process may take a very lengthy period, as is the case with Facebook's Deepface.

Figure 3.4



Feature Extraction

The face recognition system developed by Divyarajsinh N. Parmar and Brijesh B. Mehta (2013) may be divided into a few Holistic-based approaches, Feature-based methods, and Hybrid methods. In order to create a sophisticated representation for recognition, holistic-based methods—also referred to as appearance-based methods—involve the full information about a face patch. The holistic approaches PCA (Principal Component Analysis) and LDA are examples of (Linear dependent Analysis). Contrarily, feature-based approaches directly extract detail from select areas, particularly face features like the eyes, nostrils, and lips, while discarding other data that is deemed unnecessary. LBP is an instance of a feature-based approach (Local Binary Pattern). In order to boost efficiency, these approaches are frequently combined to form hybrid methods, for example the combination of the holistic-based method with the feature-based method.

4.Implementation &Testing

1.Coding Blocks

Main Frame of the System

```
import tkinter as tk
from tkinter import *
import cv2
import csv
import os
import numpy as np
from PIL import Image,ImageTk
import pandas as pd
import datetime
import time
#####Window is our Main frame of system
window = tk.Tk()
window.title("Face Attendance using Machine Learning")
window.geometry('1280x720')
window.configure(background='snow')
def manually_fill():
    global sb
    sb = tk.Tk()
    sb.title("Please enter subject Name :-")
    sb.geometry('580x320')
    sb.configure(background='snow')
```

```

def err_screen_for_subject():
    def ec_delete():
        ec.destroy()
    global ec
    ec = tk.Tk()
    ec.geometry('300x100')
    ec.title('Warning!!')
    ec.configure(background='snow')
    Label(ec, text='Please enter your subject name!!!', fg='red', bg='white', font=('times', 16, 'bold')).pack()
    Button(ec, text='OK', command=ec_delete, fg="black", bg="lawn green", width=9, height=1, activebackground="Red",
           font=('times', 15, 'bold')).place(x=90, y=50)
def fill_attendance():
    ts = time.time()
    Date = datetime.datetime.fromtimestamp(ts).strftime('%Y_%m_%d')
    timeStamp = datetime.datetime.fromtimestamp(ts).strftime('%H:%M:%S')
    Time = datetime.datetime.fromtimestamp(ts).strftime('%H:%M:%S')
    Hour, Minute, Second = timeStamp.split(":")
    #####Creatting csv of attendancc
    ##Create table for Attendance
    date_for_DB = datetime.datetime.fromtimestamp(ts).strftime('%Y_%m_%d')
    #global cursor
    global subb
    subb=SUB_ENTRY.get()

```

4.1.3 Security question code:

```
def verify_question(request):
    if request.method == "POST":
        user = request.user
        try:
            user_profile = user.userprofile # Assuming a ForeignKey relation
            security_answer = request.POST.get("security_answer")

            # Check if the provided answer matches the stored answer

            if user_profile.security_answer == security_answer:

                # Redirect the user to the home page if the answer is correct

                return redirect("home")
            else:

                # If the answer is incorrect, display an error message

                messages.error(request, "Incorrect answer to security question.")
                return redirect("login")
```

4.Implementation &Testing

4.2 Create Table For Attendance

```
date_for_DB = datetime.datetime.fromtimestamp(ts).strftime('%Y_%m_%d')
#global cursor
global subb
subb=SUB_ENTRY.get()
DB_table_name = str(subb + "_" + Date + "_Time_" + Hour + "_" + Minute + "_" + Second)
import pymysql.connections
global cursor
###Connect to the database
try:
    global mycursor
    connection = pymysql.connect(host='localhost', user='root', password='root', db='manually_fill_attendance')
    mycursor = connection.cursor()
except Exception as e:
    print(e)
sql = "CREATE TABLE " + DB_table_name + ""
      (ID INT NOT NULL AUTO_INCREMENT,
      ENROLLMENT varchar(100) NOT NULL,
      NAME VARCHAR(50) NOT NULL,
      DATE VARCHAR(20) NOT NULL,
      TIME VARCHAR(20) NOT NULL,
```

```
print(ex) #
    if subb=="":
        err_screen_for_subject()
    else:
        sb.destroy()
        MFW = tk.Tk()
        #MFW.title("Manually attendance of "+ str(subb))
        MFW.geometry('880x470')
        MFW.configure(background='snow')
        def del_errsc2():
            errsc2.destroy()
        def err_screen1():
            global errsc2
            errsc2 = tk.Tk()
            errsc2.geometry('330x100')
            errsc2.title('Warning!!')
            errsc2.configure(background='snow')
            Label(errsc2, text='Please enter Student & Enrollment!!!', fg='red', bg='white',
                    font=('times', 16, 'bold')).pack()
            Button(errsc2, text='OK', command=del_errsc2, fg="black", bg="lawn green", width=9, height=1,
                    activebackground="Red", font=('times', 15, 'bold')).place(x=90, y=50)
        def testVal(inStr, acttyp):
            if acttyp == '1': # insert
                if not inStr.isdigit():
```

```
return False
```

```
    return True
```

```
ENR = tk.Label(MFW, text="Enter Enrollment", width=15, height=2, fg="white", bg="blue2",  
               font=('times', 15, ' bold '))
```

```
ENR.place(x=30, y=100)
```

```
STU_NAME = tk.Label(MFW, text="Enter Student name", width=15, height=2, fg="white", bg="blue2",  
                    font=('times', 15, ' bold '))
```

```
STU_NAME.place(x=30, y=200)
```

```
global ENR_ENTRY
```

```
ENR_ENTRY = tk.Entry(MFW, width=20, validate='key', bg="yellow", fg="red", font=('times', 23, ' bold '))
```

```
ENR_ENTRY['validatecommand'] = (ENR_ENTRY.register(testVal), '%P', '%d')
```

```
ENR_ENTRY.place(x=290, y=105)
```

```
def remove_enr():
```

```
    ENR_ENTRY.delete(first=0, last=22)
```

```
STUDENT_ENTRY = tk.Entry(MFW, width=20, bg="yellow", fg="red", font=('times', 23, ' bold '))
```

```
STUDENT_ENTRY.place(x=290, y=205)
```

```
def remove_student():
```

```
    STUDENT_ENTRY.delete(first=0, last=22)
```

```

def enter_data_DB():
    ENROLLMENT = ENR_ENTRY.get()
    STUDENT = STUDENT_ENTRY.get()
    if ENROLLMENT=="":
        err_screen1()
    elif STUDENT=="":
        err_screen1()
    else:
        time = datetime.datetime.fromtimestamp(ts).strftime('%H:%M:%S')
        Hour, Minute, Second = time.split(":")
        Insert_data = "INSERT INTO " + DB_table_name + " (ID,ENROLLMENT,NAME,DATE,TIME) VALUES (0, %s,
%s, %s,%s)"
        VALUES = (str(ENROLLMENT), str(STUDENT), str(Date), str(time))
        try:
            mycursor.execute(Insert_data, VALUES)
        except Exception as e:
            print(e)
        ENR_ENTRY.delete(first=0, last=22)
        STUDENT_ENTRY.delete(first=0, last=22)

def create_csv():
    import csv
    mycursor.execute("select * from " + DB_table_name + ";")
    csv_name='./Manually Attendance/'+DB_table_name+'.csv'
    with open(csv_name, "w") as csv_file:
        csv_writer = csv.writer(csv_file)

```

4.Implementation &Testing

User Registration: Allows administrators to register new users (students, teachers, administrators) into the system, capturing their biometric data (facial images) and relevant information.

User Authentication: Verifies the identity of users accessing the system through login credentials (username and password) or biometric authentication (facial recognition).

Attendance Recording: Automatically records student attendance based on facial recognition during class sessions or events, eliminating the need for manual roll calls.

Attendance Monitoring: Provides real-time monitoring of student attendance status, allowing teachers and administrators to track attendance trends and identify patterns.

Database Management: Manages the storage and retrieval of student information, including facial images, attendance records, and other relevant data.

Face Detection: Identifies and detects faces in images captured by the system's camera, utilizing algorithms such as Viola-Jones or Convolutional Neural Networks (CNNs).

Face Recognition: Matches detected faces against the stored database of registered students, accurately identifying individuals and marking their attendance.

User Interface (UI): Provides a user-friendly interface for administrators, teachers, and students to interact with the system, featuring functionalities such as user registration, attendance tracking, and database management.

Data Analysis and Reporting: Analyzes attendance data to generate comprehensive reports and insights, facilitating administrative decision-making and compliance reporting.

Security and Access Control: Implements robust security measures

Conclusion and Results

1.Resulting Screens

Figure 5.1

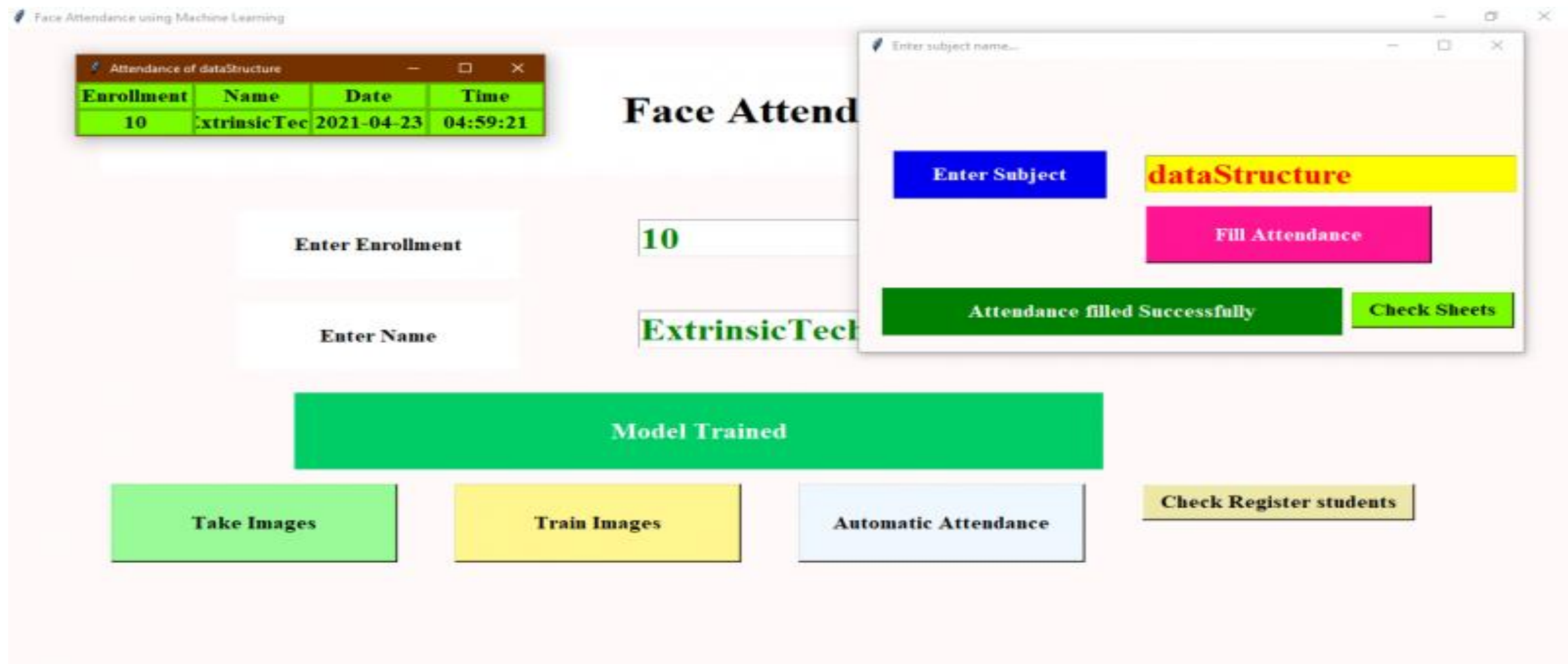


Figure 5.2

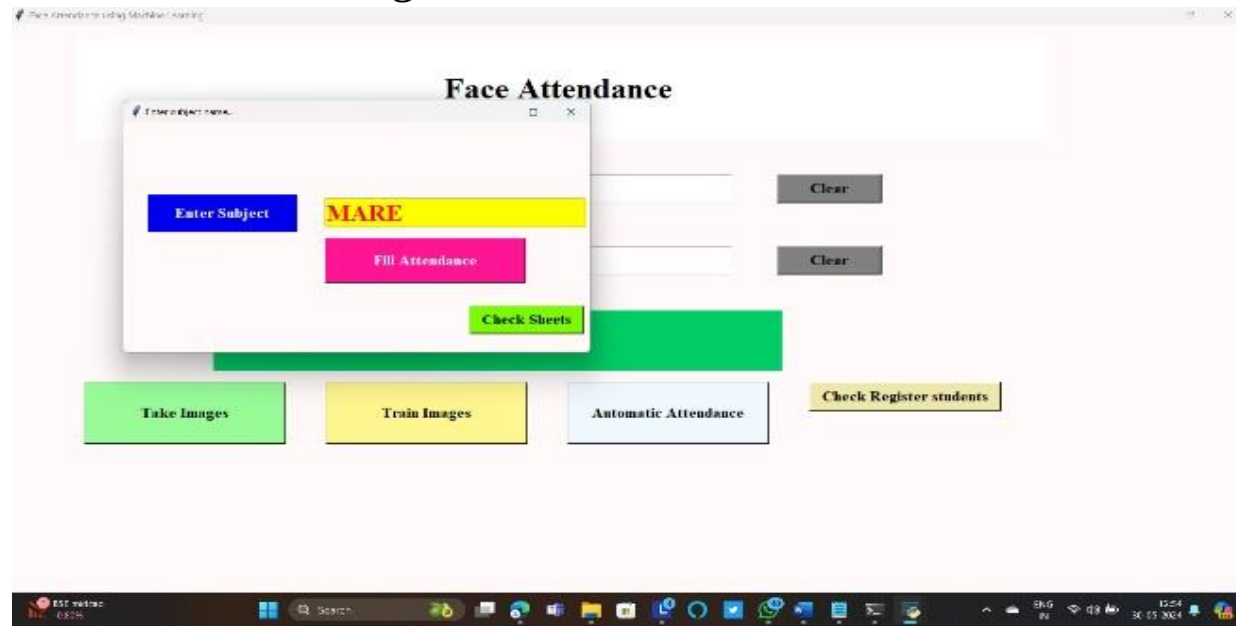


Figure 5.3

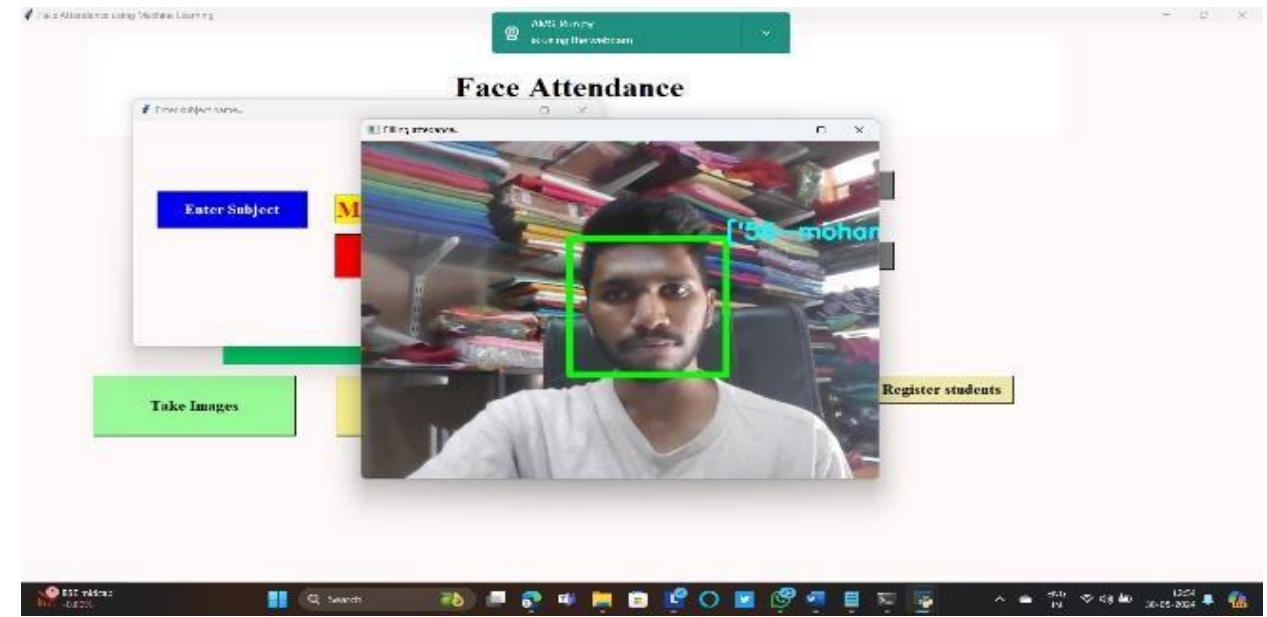


Figure 5.4



CONCLUSION:

This method provides a detailed description of an automated student attendance system based on facial recognition. With the help of a comparison between the input picture from a video frame that was recorded and the training image, the suggested strategy offers a way to recognize the persons. This suggested method is capable of locating a face from an input facial picture that is taken from a frame of the recorded video. Moreover, it offers a pre-processing technique to improve image contrast and lessen the lighting impact. Both LBP and PCA are used to extract characteristics from the face picture. The system may be stabilized by producing consistent results thanks to the method that combines LBP and PCA. When two photographs per person are taken, the accuracy of the suggested method is 100% for high-quality images, 92.31% for low-quality images, and 95.76% for the Yale face database.

FUTURE SCOPE:

There are certain restrictions in the suggested technique. The input image must first be frontal and a single, upright face image. Second, if there are severe lighting issues, accuracy may suffer. Thirdly, if the image that was obtained is blurry, incorrect recognition could happen. Moreover, LBP is a descriptor that extracts local characteristics based on textural data. To achieve high accuracy, test and train images must be of the same quality and be taken with the same instrument. Finally, if a person is wearing makeup in a face recognition photo, the key characteristics will be hidden.

A better camera with a stronger light source can lessen the illumination issue and prevent the acquisition of blurry photographs. With the suggested method, the laptop's built-in camera serves as the default device. The laptop camera's illumination, however, is quite weak, which makes the system unsteady. To have better results in the future, a better camera and lighting source can be employed. This can lessen the reliance on ambient brightness, particularly in locations where test and training photos are taken. Moreover, it is possible to construct a face recognition system that recognizes several faces in addition to a single facial picture. This may improve the system's effectiveness. A technique, such as the affine transform, can be used in the pre-processing step to align the face picture based on coordinates in the center of the eyes. This might be beneficial, particularly for the PCA technique, which maps the test picture to the training image to do face recognition. This approach largely depends on the camera that was used to take the photographs; therefore the test image and train image are related to one another. This tactic must make use of the same capturing tools in order to be successful. As a result, alternative algorithms, such as the A.I. (artificial intelligence) algorithm, which may be used for face identification, may take the role of LBP. Convolution Neural Networks (CNNs), a machine deep learning technology that has recently gained popularity, may do recognition with less reliance on a particular train image given a large library. CNN, on the other hand, requires either a very small class size or a very large database to increase its accuracy.

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THANK YOU